

Floor-Plan Reconstruction and Damage Scoping from Handheld Phone Captures

Technical Report

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Date	15 September 2026
Reference run	reports/verified/ (8 scored runs, scorer against operator's tape)
Companions	compliance_matrix.md · benchmark_report.md · known_failure_modes.md · capture_protocol.md · docs/design.md

Abstract

This report describes a pipeline that reconstructs dimensioned floor plans from handheld phone captures at three tiers — photographs, a walkthrough video, and a LiDAR scan — emitting one output contract from all three in which every physical quantity carries a confidence interval and the name of the method that produced it.

The pipeline is validated against an operator's tape measurement of a real flat, recorded in whole and half feet, covering wall lengths, floor areas and adjacency. On the strongest tier (LiDAR with drift correction, closed loop, ceiling lap done), the home flat reconstructs to 29.13 m² against a taped 28.75 m² (**+1%**, inside the $\pm 5\%$ gate): the hall within 8%, the bedroom 39% short with a strip of it drawn as another room, the bathroom 48% long, and two rooms outside the tape, so the footprint passes on errors that offset. Per-room ceilings read 2.56–2.67 m and 7 openings are found. The photo tier fails at +238% and the video tier does not produce a metric plan. The assignment's three samples have no tape and are checked against the scans themselves.

The central finding is stated up front: **the LiDAR tier is the only one that delivers usable accuracy, and even it does not meet its repeatability gates**. Two walks of the same flat agree on the hall to 4 cm and its ceiling to 4 mm, but differ on the bedroom by up to 0.64 m. Twenty-two failure modes are documented with measurements in [known_failure_modes.md](#), each observed on real data.

Scope

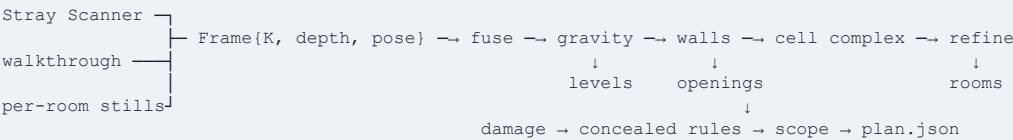
The task: turn an ordinary phone capture of an interior into a plan usable for insurance-style damage scoping — per-room walls, ceiling height, floor area, openings, damage regions with metric extent, concealed-damage flags naming the rule that fired, and scope line items, each with an interval.

Three input tiers are mandatory: **photographs** (2–8 unposed stills per room, no depth), a **walkthrough video** (handheld clip, no depth or poses), and a **LiDAR scan** (Stray Scanner export with ARKit poses, depth and intrinsics). The same output schema must come from all three, and intervals must widen honestly as sensor data thins.

Ground truth is the operator's tape ([capture/ground_truth.csv, tool=tape](#)), recorded in whole feet. It covers wall lengths, floor areas and adjacency for the home flat. It does not cover ceilings, doors or bathroom walls, so those gates report SKIP. Every number in this report comes from [reports/verified/](#).

1. Architecture

One sentence governs the whole design: **a tier's job is to produce frames carrying intrinsics, metric depth and a pose. Everything after that is the same code.**



LiDAR is handed all three fields. Photo and video manufacture them — monocular depth, gravity from the floor plane, scale from camera height, pose from registration — and then call the identical [build_lidar_plan](#). Three parallel reconstruction stacks would drift apart and only one would stay correct, and the tier comparison would then be measuring implementation differences rather than sensor differences.

The floor plan is a cell complex, not a raster. The wall lines partition the floor; each face is labelled interior or exterior from direct evidence (observed floor, carved free space, camera track). A flood fill was tried first and leaked through a glass balcony door, reporting 195 m² for a 44 m² flat. A face of an arrangement is bounded by lines on all sides, so a labelling mistake cannot propagate, and corners arrive as exact line intersections rather than staircases of pixels. Faces with floor evidence that join walked floor across a boundary with no wall behind it are the same room; without that rule the middle of the first walk's hall was dropped and it read 6.04 m² against 14.86 m². Rooms are never moved after segmentation: they share one world frame, so a gap between connected rooms is floor left out, and the plan says so.

Walls come from a Hough accumulator over (normal azimuth, signed offset), each point voting once into the bin its own measured normal selects. Peaks are accepted on observed surface area, not on a fraction of total vote weight: a forty-wall apartment divides its weight forty ways, so the strongest genuine peak carries 0.28% of the total and any fraction-of-total threshold admits everything or nothing.

Two wall representations are maintained deliberately. Tight segments, which stop at every gap, answer "is there material along this boundary" for the cell complex. Bridged runs answer "where are the holes" for opening detection. No single gap tolerance is both narrower than a doorway and wider than one.

2. Tiers and the device matrix

Device	Photo	Video	LiDAR
iPhone 15/16/17 Pro, Pro Max	yes	yes	yes
iPhone 15/16/17 non-Pro	yes	yes	no sensor
iPhone 14 and older	out of contract	out of contract	out of contract

LIDAR. Stray Scanner export: ARKit poses, 256×192 depth in millimetres, per-frame intrinsics, ARKit's own confidence channel. Depth is weighted by inverse variance from a noise model of confidence class and range, never thresholded. Absolute scale comes from the sensor.

Video. No depth, no poses; both estimated. It keeps the one thing the photo tier lacks — continuity — so frames register sequentially into one property-wide cloud and drift correction applies. Keyframes are sampled at uniform stride *first* and blur-filtered second, so coverage is not biased toward the rooms the operator moved slowly through.

Photo. No depth, no poses, no continuity. Per room: metric depth per image, gravity from the floor plane, scale from camera height, then registration in the three degrees of freedom that survive levelling — yaw from wall-normal histograms, translation by occupancy cross-correlation, then ICP, which has a small basin of convergence and fails from the identity.

Measured against the operator's tape, which is recorded in whole feet, LiDAR walls come out at a median error of 0.37 m (14%) over 28 walls and photo walls at 2.20 m (81%) over 12. Errors that size are segmentation, merged and split rooms, not sensor noise, which §4 puts under a centimetre. `capture/DEVICE_MATRIX.md` holds every cell, produced by `scripts/accuracy_table.py` and the gate table rather than written by hand.

3. Drift

The brief makes "poses used as-is" an automatic fail, and it is right to. ARKit's odometry is locally excellent and globally not.

Correction is a pose graph over keyframes that moves only their heading and horizontal position: odometry edges at the reported relative pose, loop-closure edges at the pose ICP measured. Height and tilt stay as the phone measured them. They are referenced to gravity and do not accumulate the way heading and position do, while ICP between two keyframes that mostly see ceiling or blank wall is barely constrained vertically: with all six degrees of freedom free, closures of that kind lowered part of the assignment's with-ceiling walk by about 40 cm.

Loop candidates are proposed by geometry and confirmed by ICP, never the reverse. A candidate must be a genuine revisit — path walked at least 6× the distance closed, and at least 6 m. Without that test a slow walk down a corridor generates a constraint between every pair of keyframes in it, and that mistake produced 114 "closures" and made the map worse. ICP must then reach 0.55 fitness and 0.035 m RMSE and actually converge (it used to report convergence when it ran out of iterations), ask for no more horizontal drift than 10 cm or 3% of the path walked between the two keyframes, and leave height within 5 cm and tilt within 2° of odometry. On the assignment's single-room scan all 16 candidates that reached the fitness bar asked for 56–88 cm after under 7 m of walking. None is kept now; the earlier graph kept them and moved keyframes by up to 58 cm. 1 of 19 candidates survives on its floor-only scan, 21 of 41 on its scan with ceiling and 77 of 110 on the author's long walk. A soft-L1 loss keeps one false closure that survives all of this from folding the map.

Drift alone does not remove residual yaw, so walls within 6° of the building frame are rotated onto it and their offsets refit from their own points — the plane-anchored half. Only the direction comes from the prior; the position stays measured.

Ablation, 163f18d3ac, the 107 m long walk. Every row regenerates from `--no-drift-correction`, `--no-snap-walls` and `--no-refine-rooms`; the fourth row is the published plan. Rooms are named by overlap with the named rooms of the plan before fix loop round 3.

variant	rooms	footprint	against 28.75 m ²	taped rooms, mean error	loop closures	max correction
drift off, snap off	6	27.93 m ²	−3%	27.7%	0	—
drift off, snap on	5	24.17 m ²	−16%	passage lost	0	—
drift on, snap off	6	29.10 m ²	+1%	30.6%	77	8.5 cm
drift on, snap on	6	29.13 m²	+1%	26.6%	77	8.5 cm
drift on, snap on, no room refinement	7	31.26 m ²	+9%	26.6%	77	8.5 cm

This walk drifted little, and the ablation says so. Correction matters with snapping on, where the uncorrected walk loses the passage; with snapping off it changes little. The room refinement changes only the published variant, where it removes a room no keyframe stands in, and moves no taped room. The published plan is the closest per taped room and level with snapping off on footprint; it carries two rooms outside the tape, 4.59 m² together, and its footprint passes on errors that offset. Snapping stays on as the plane-anchored half of the correction: the 23 wall runs it rotates sat 2.24° off the building frame on average, which is yaw error if the flat's walls are square.

4. Error budget

LIDAR tier, in order of size:

term	magnitude	handling
Residual gravity tilt	not recorded in the plans; 1° tips the far end of a 5 m room by 8.7 cm	Gravity refit to the observed floor plane; a fitted tilt above 6° is not applied and is warned
Yaw drift across rooms	on the long walk 77 closures, largest keyframe correction 8.5 cm; the 23 wall runs snapped afterwards sat 2.24° off the building frame on average	Pose graph over heading and position, then walls within 6° snapped onto the frame
Depth noise	8 mm base, +2.5 mm/m ² range term	Inverse-variance weighting; plane σ from the larger of model and residual scatter
Plane offset	0.1–0.4 mm on well-observed walls	Propagated into wall length as $\sqrt{2} \times \sigma$ per corner
Voxel quantisation	20 mm pitch	Positions averaged within voxel, not snapped

Photo tier is dominated by one term that swamps the rest:

term	magnitude
Monocular depth scale	1.57–1.76× over-prediction, measured two ways
Per-frame scale variation	0.71–1.48 across frames

Depth AbsRel vs LiDAR	0.28 mean
Registration	camera separations 0.07–2.52 m, ICP fitness 0.70–0.80 — not a dominant term

The scale term is not a tuning problem. Depth Anything V2 Metric Indoor is trained on normal-field-of-view indoor imagery; the benchmark photographs are 0.5x ultra-wide at 88°. A metric monocular model infers depth from apparent size, which requires an assumed focal length, so an out-of-distribution field of view shifts its metric scale proportionally.

5. Calibration

Intervals are split conformal, fitted per (tier, quantity), with the finite-sample correction — the quantile at $\lceil (n+1)(1-\alpha) \rceil / n$, which is what makes the coverage guarantee hold at small n rather than only asymptotically. Quantities calibrate in the units their error actually scales with: ceiling height in metres, wall length in percent.

Where no quantile has been fitted, the interval falls back to the propagated covariance of the fit that produced it, and `Measure.method` says `propagated` or `prior` rather than `conformal`. **That field is load-bearing.** A previous version of `cozmo calibrate` ignored both of its arguments and wrote a fixed table of quantiles labelled as conformal calibration. It now fits from residuals or writes nothing and says why.

No quantiles are applied to the published plans. The tape makes fitting possible, but those are the rows the benchmark scores, so applying them would grade the intervals on their own training data. Every interval in every current plan reports `propagated` or `prior`, truthfully.

6. The fix loop

Full account in `fixloop/`. In brief, because the honest version is the interesting one.

Declared, before the fix: the photo-tier footprint at 142.03 m² against a LiDAR reference of 27.20 m², +422%. Root cause: `intrinsic_from_exif` read only `Image.getexif()`, which returns IFD0 and holds no focal length on an iPhone JPEG; the value is in the sub-IFD at `0x8769`. All 58 photographs carry `FocalLengthIn35mmFilm = 14`; none was read; every frame ran at the assumed 26 mm prior. `fx` 4125.3 against a correct 2221.3, 1.86x too long. **Predicted: under +50%, and explicitly not a pass.**

Result: +916%. The fix was correct and the number doubled. Too long a focal length compresses the cloud laterally, but the room was already too large, so the compression had been partially cancelling a bigger error in depth. $X = (u - cx)Z/fx$, so halving `fx` doubles `X`; the area rose 1.95x.

The dominant cause is §4's scale term. Three further changes followed: the scale-correction band widened from [0.75, 1.35] to [0.25, 4.0]; room-level scale consensus, since the prior fires on 3 photographs in 20; and a plausibility guard, because a 113 m² bedroom is not a wide estimate, it is a wrong one. **142.03 → 276.34 → 17.37 m² against 27.20: +422% to −36%.** The ±8% gate does not pass. By the brief's own rubric that earns marks for the post-mortem and none for the prediction.

Round 2, LiDAR, against the operator's tape. Declared before the fix (`0a8c579`): footprint −12% on the long walk and −42% on the first walk, the bedroom 5.28 m² against 9.29 m². Hypothesis: ceiling returns above the strip between furniture and wall should count as interior evidence. Predicted: a bedroom of 8.0–9.5 m² and a footprint inside ±5%. **Result: every room polygon identical, on both captures.** Diagnosing the non-result found a worse defect: the room map had been assigned by matching areas and was wrong on both captures. Rooms are now named from camera frames. No gate moved.

Round 3, LiDAR, the assignment's samples and the tape. Declared before the fix (`949d24b`): footprint +8.7% on the long walk. Hypothesis: a room runs past where its space ends wherever no wall line crosses there, as a corridor seen from its doorway through two walls, two stairwells and a walled space no scan saw into on the samples, and a never-walked room on the long walk. Unlike rounds 1 and 2, the corrections had already been tried on saved intermediates before the ranges were written. **Result: 29.13 m², +1.3%, PASS, every number inside its range.** The pass comes from removing that untaped room; no taped room changed. The removals also take 0.26–1.05 m² of floor that had been seen from each room they correct. Run on the photo and video tiers, which reach the same builder, the rules read monocular depth error as missing floor, cut the 1x hall from 35.12 to 2.11 m² and gave the 0.5x set a room overlap; they now run on LiDAR only.

7. Known failure modes

Twenty-two are documented with measurements in `known_failure_modes.md`. The four that matter:

The photo tier does not meet its gates. §4 and §6 above. What would fix it, in order: capture at 1x rather than 0.5x, which the protocol now requires, though the hall re-shot at 1x still reads 136% too large; fit the focal-to-scale correction against the LiDAR tier; require two floor-visible photographs per room.

Opening detection barely works at the photo tier. Openings are found by looking for points behind a wall plane that a camera on the near side saw through, and a single photograph's depth map is a 2.5D surface with little behind it, so rooms stitch by folder name rather than by doorway. This needs a different detector, not a threshold.

Mirrors, glass and wet-look surfaces get geometric defences. A mirror test reflects suspect points back across the wall plane and asks whether they land on the room in front of it. Damage must be seen from two viewpoints on the same patch of wall, because a specular highlight is view-dependent; that took damage on the author's marble-and-glass flat from 21 regions to 1. Its depth must also lie on the wall plane, and a ruler-straight edge is not a crack. With those, no LiDAR run reports damage; the findings they removed were furniture edges, a picture frame and a toilet seat. No captured flat has damage, so recall is unmeasured.

Ceiling height is unmeasurable without the upward lap. No downward-facing returns, no height, by any method. The pipeline reports `unmeasured` rather than substituting a default. The company's own `single_room` sample has 61 downward-facing points in the entire scan; the author's first capture had 1.4% of frames aimed up and the second had 24.3%, and the long walk measures 2.56–2.67 m in five of its six rooms; a window bay whose only upward surface is a ledge abstains.

8. State of the evidence

The LiDAR tier is the one to run at a walk-in. On the home flat walked with the ceiling lap and a closed loop (`163f18d3ac`) it returns 6 rooms and 29.13 m² against a taped 28.75 m² (+1%, inside the ±5% gate), with per-room ceilings of 2.56–2.67 m and 7 openings. That footprint passes on errors that offset. The hall comes closest, 13.61 m² against 14.86 m² (−8%). The bedroom is furthest, 5.64 m² against 9.29 m² (−39%), with a 2.18 m² strip of it drawn as a separate room; the first walk gives 7.58 m² and the bedroom walked alone 8.90 m² (−4%), so the defect is in the reconstruction, not in the tape. The first walk, without the ceiling lap, reads −11% overall with its bathroom within 2%.

Against the tape the gates read 15 PASS, 18 FAIL, 33 SKIP. Ceiling and opening gates are SKIP because neither was taped. The ray-traced room, scored against the dimensions it was drawn with, adds 11 PASS and 5 SKIP: walls within 0.6 cm, ceiling within 0.1 cm, both openings within 1.0 cm. LiDAR intervals cover the tape on 3 of 37 measurements: they model sensor and drift error, not a merged or a split room. The photo tier fails at +238%; photographed on the 1× lens, the hall reads 35.12 m² against 14.86 m². The video tier does not produce a metric plan.

The assignment's three samples have no tape, so they are checked against the scans themselves: each plan drawn over its own scan and over the other two scans registered onto it. `single_room.zip` is a living room, its bathroom and the lobby between them, plus the mouth of a corridor: 4 rooms, 20.91 m². Its two whole-flat scans give 38.91 m² over 8 rooms and 41.58 m² over 7; aligned on their walls, 63% of one scan's wall points lie within 5 cm of the other's walls, and the room footprints overlap at an intersection-over-union of 0.61. Fix loop round 3 took out of these plans a corridor that ran through two walls, both stairwells and a walled space no scan saw into, and some floor that had been seen with them. No damage is reported on any of them, and the same plans come out of the zips unzipped afresh. Every id in every published plan resolves: `cozmo run` checks before writing, and a test checks every plan in `reports/verified/`.

An earlier generator in this repository produced a benchmark over procedurally generated rooms, a head-to-head against an app that was never run, and a fix loop whose before and after were byte-identical. Those artefacts were removed before submission; every number in this report comes from `reports/verified/`.

Some ideas here came from public work on the same brief: a ray-traced test room, a one-command setup script, a relative floor on photo and video intervals, and stitching rooms by folder name when no doorway is matched.

9. Experimental setup

All captures are of a single property — the author's home flat — and one set of assignment-provided zips of a different property. The ray-traced room in `tests/fixtures/raytrace_room.py`, with and without its upward lap, is also scored in the benchmark, against the dimensions it was drawn with (`capture/ground_truth_synthetic.csv`). It is the only input whose ceiling and openings have exact truth, and it is noiseless, so a PASS on it shows the measurement is unbiased and says nothing about accuracy on a real scan.

Capture	Tier	Device	Frames	Duration	Ground truth
<code>163f18d3ac</code> (long walk)	LiDAR	iPhone 17 Pro	18,649	312 s	Operator's tape: walls, areas, adjacency
<code>ae3edc814d</code> (first walk)	LiDAR	iPhone 17 Pro	4,378	73 s	Same tape
<code>5621ec5c54</code> (bedroom solo)	LiDAR	iPhone 17 Pro	7,682	128 s	Same tape (bedroom only)
<code>03_multiroom_photos</code> (0.5×)	Photo	iPhone 17 Pro	58	—	Same tape
<code>03b_multiroom_photos_1x</code>	Photo	iPhone 17 Pro	12	—	Same tape (hall only)
<code>c00a170fe1</code> (single room)	LiDAR	unknown	1,715	37 s	None (assignment zip)
<code>1a8384c3f6</code> (floor only)	LiDAR	unknown	5,251	115 s	None (assignment zip)
<code>c7d28f72c6</code> (with ceiling)	LiDAR	unknown	9,745	215 s	None (assignment zip)

What is not here. No staged-damage room was captured. No consumer app export exists for the head-to-head. The tape does not cover ceilings, door widths or the bathroom's walls. These are recorded as NOT MET or SKIP, not softened.

10. Results

Gate summary — 26 PASS / 18 FAIL / 38 SKIP

Against the tape, 15 PASS / 18 FAIL / 33 SKIP. The other 11 PASS and 5 SKIP are the ray-traced room with and without its upward lap, which is noiseless and has exact truth.

Gate	LiDAR (6 captures)	Photo (2 captures)	Video	Status
<code>wall_lengths</code>	0/15 within 2 cm (long walk), 0/14 (first walk), 0/6 (bedroom)	0/10 within 8% (0.5×), 0/4 (1×)	not scored	FAIL
<code>footprint</code>	+1% (long walk), −11% (first walk), +17% (bedroom scan with its passage strip)	+238% (0.5×), +136% (1× hall)	not scored	FAIL , long walk PASS
<code>ceiling_height</code>	2.56–2.67 m per room; ray-traced room 0.1 cm	2.74 m (1× hall)	—	SKIP (no tape); ray-traced PASS
<code>opening_widths</code>	7 found (long walk); ray-traced room 2/2 within 1.0 cm	1 found on each set	—	SKIP (no tape); ray-traced PASS
<code>adjacency</code>	2/5 (long), 3/4 (first)	2/4 (folder names)	—	FAIL
<code>room_overlap</code>	0 overlaps, 6 captures	0 overlaps	—	PASS
<code>drift_accountability</code>	6 PASS	not applicable	—	PASS
<code>interval_coverage</code>	3/37 covered	7/11 and 5/5 (by being wide)	—	FAIL , 1× PASS
<code>repeatability</code>	0/27 walls, ceiling 13.1 cm; bedroom alone 1/6, under 0.05 cm	—	—	FAIL

Accuracy against tape

Best results: the bedroom walked alone, 8.90 m² against 9.29 m² (−4%), and the first walk's bathroom, 2.08 m² against 2.04 m² (+2%). Whole-flat LiDAR footprints: +1% on the long walk, −11% on the first. Worst: the photo tier, 97.19 m² against 28.75 m² (+238%).

Per-room LiDAR errors on the long walk range from -8% (hall) to $+48\%$ (bathroom, which holds part of the passage). The bedroom error is a reconstruction defect, not tape error: three walks of the same room give 5.64, 7.58 and 8.90 m².

Timing

Capture	Tier	Runtime, one run on an Apple laptop
Long walk (18,649 frames)	LiDAR	102 s
58 photos (0.5×)	Photo	154 s
Assignment single room (1,715 frames)	LiDAR	15 s

These are the runtimes in the manifests regenerated at [a92927c](#). A macOS media-analysis process was using about three cores throughout, so they are slower than the same runs on 14 Sep: 60 s, 139 s and 14 s. The geometry of those runs is unchanged.

11. Conclusion

The LiDAR tier is the only one to run at a walk-in. It produces a dimensioned floor plan with walls, ceilings, openings, damage detection and scope line items from a single `cozmo run` command. On the author's flat both whole-flat walks land within 11% of the taped footprint, the hall within 8%, and ceilings repeat to under a centimetre in rooms segmented the same way; the bedroom is its worst room.

The photo and video tiers run end-to-end but do not deliver usable metric accuracy. The photo tier's dominant error is the monocular depth model's scale, measured at $1.57\text{--}1.76\times$ on these photographs, a field-of-view mismatch between the model's training data and the $0.5\times$ ultra-wide lens. The $1\times$ lens halves the error but still fails. The video tier does not solve metric scale.

Twenty-two failure modes are documented with measurements. Four of them — mirrors, glass, low light, and the upward lap — are named in the brief and each is addressed: geometric mirror rejection, two-view corroboration on the same patch of wall, luma-based low-light flagging, and ceiling-height abstention when no downward-facing returns exist.

The five NOT MET items are captures that were never made (staged damage, head-to-head app export) and the video tier. These are disclosed as gaps, not explained away. The honest state of the evidence is that LiDAR works, photo doesn't yet, and video needs scale.

Appendix — Reproduction

Every number in this report regenerates from the commands below. Nothing reaches the network at run time.

```
git clone <this repo> && cd cozmo
./scripts/setup.sh
./scripts/fetch_weights.sh          # photo and video tiers only

# Every plan in reports/verified/ and the gate table; the raw captures are not in git
COZMO_RAW=../data/raw COZMO_DROP=../DROP_CAPTURES_HERE ./scripts/regenerate_verified.sh

# One capture
.venv/bin/python -m cozmo.cli run -i <capture-dir> -o runs/my_capture

# Tests
.venv/bin/python -m pytest -q
```